

SANTIAGO DEL ESTERO, Argentina

WMO: 871290

Lat: 27.7547S

Lon: 64.2997W

Elev: 653

StdP: 14.35

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	30.9	34.8	18.3	14.2	45.2	22.0	16.9	47.7	22.1	64.5	19.9	69.5	1.8	320	0.363

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.8	102.6	74.4	99.8	74.2	96.9	73.7	79.6	92.7	78.2	91.5	77.0	90.2	10.6	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.2	140.3	86.4	74.8	133.8	84.4	73.5	127.8	83.0	43.8	93.0	42.3	91.6	41.0	90.3	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.5	17.9	14.5	DB	24.7	107.7	3.0	2.3	22.6	109.3	20.8	110.7	19.1	112.0	16.9	113.6	(4)
(5)			WB	22.9	81.8	3.1	1.7	20.7	83.0	18.9	84.0	17.2	85.0	14.9	86.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	69.7	81.2	79.1	75.3	69.0	62.1	56.5	55.7	60.6	66.5	73.4	77.0	80.2	(6)
(7)		DBStd	11.28	5.78	6.30	5.96	6.93	6.91	6.27	7.29	8.28	8.31	7.46	6.81	6.21	(7)
(8)		HDD50	49	0	0	0	0	2	14	25	7	1	0	0	0	(8)
(9)		HDD65	1028	0	1	4	36	141	261	300	189	80	13	3	0	(9)
(10)		CDD50	7230	967	815	785	571	378	210	202	335	495	724	811	937	(10)
(11)		CDD65	2736	502	396	323	158	52	8	12	52	124	273	364	472	(11)
(12)		CDH74	30872	6018	4315	3021	1282	367	59	200	809	1638	3161	4310	5692	(12)
(13)	CDH80	15314	3268	2168	1320	448	82	6	49	333	717	1573	2257	3093	(13)	
(14)	Wind	WSAvg	5.8	5.7	5.5	5.1	5.1	4.7	4.6	5.3	6.0	6.8	7.2	6.9	6.5	(14)
(15)	Precipitation	PrecAvg	23.3	4.8	4.1	3.4	1.4	0.6	0.3	0.1	0.0	0.4	1.7	2.7	3.8	(15)
(16)		PrecMax	45.8	12.7	20.6	8.0	7.4	2.4	1.9	0.9	0.4	2.8	8.3	10.1	11.7	(16)
(17)		PrecMin	11.1	0.7	0.3	0.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	(17)
(18)		PrecStd	6.3	2.7	3.1	1.9	1.3	0.6	0.4	0.2	0.1	0.6	1.7	1.8	2.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	105.2	103.5	99.3	93.4	86.6	80.6	86.1	96.0	100.2	103.8	104.7	106.6	(19)
(20)			MCWB	75.6	75.4	75.5	73.6	70.7	67.4	65.1	68.0	69.1	72.8	74.6	74.3	(20)
(21)		2%	DB	100.9	98.8	94.5	89.1	82.0	75.0	78.9	88.3	93.1	98.2	99.5	101.9	(21)
(22)			MCWB	75.6	76.1	75.3	72.1	68.5	63.0	63.2	65.9	67.3	71.5	72.6	74.8	(22)
(23)		5%	DB	97.9	95.3	91.0	85.2	78.0	71.6	74.8	82.6	88.1	93.3	95.8	97.9	(23)
(24)			MCWB	74.8	75.8	74.3	71.6	66.4	60.2	60.1	62.6	64.9	69.5	71.5	74.4	(24)
(25)	10%	DB	94.5	91.7	87.4	81.2	74.3	68.4	70.4	77.3	83.5	88.9	91.9	94.1	(25)	
(26)		MCWB	74.4	74.7	72.9	69.4	64.4	58.7	56.7	59.8	62.9	67.9	70.5	73.6	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.0	81.4	79.7	77.3	72.9	68.8	67.5	70.0	72.8	76.9	80.1	80.9	(27)
(28)			MCD B	94.0	94.4	91.2	87.7	83.0	76.8	82.5	89.6	93.3	96.5	91.7	95.3	(28)
(29)		2%	WB	79.1	79.1	77.6	75.2	70.6	66.0	64.6	67.0	69.8	74.4	76.4	78.3	(29)
(30)			MCD B	93.1	91.4	89.0	84.4	78.0	72.3	76.6	85.9	87.1	91.9	91.7	93.9	(30)
(31)		5%	WB	77.6	77.7	76.2	73.3	68.6	63.6	62.2	64.3	67.5	72.1	74.4	76.8	(31)
(32)			MCD B	91.4	89.8	86.9	81.1	75.0	68.4	72.0	80.2	84.1	87.6	90.1	91.7	(32)
(33)	10%	WB	76.3	76.3	74.6	71.5	66.7	61.2	59.3	61.2	64.9	70.1	72.5	75.4	(33)	
(34)		MCD B	89.4	87.9	83.9	78.5	72.8	65.9	67.1	74.8	79.9	84.4	86.9	89.2	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	21.8	20.1	19.0	19.1	19.5	21.0	25.4	28.0	27.1	23.6	23.3	22.3	(35)
(36)			MCDBR	26.9	25.3	25.0	24.7	24.6	26.7	31.7	35.2	33.6	31.0	29.1	28.6	(36)
(37)			MCWBR	8.7	8.2	9.7	11.4	12.8	15.9	18.0	17.2	15.1	12.2	10.7	9.6	(37)
(38)		5% WB	MCDBR	22.4	21.1	19.8	18.8	18.9	18.4	26.8	32.3	28.5	25.9	23.6	23.2	(38)
(39)			MCWBR	8.9	8.5	8.9	9.8	10.3	12.1	15.8	16.0	13.6	11.2	10.1	9.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.410	0.407	0.393	0.380	0.354	0.339	0.341	0.396	0.462	0.442	0.411	0.412	(40)	
(41)		taud	2.396	2.405	2.443	2.436	2.461	2.481	2.441	2.267	2.077	2.219	2.339	2.375	(41)	
(42)		Ebn at Noon	296	292	286	273	266	264	268	264	260	278	294	296	(42)	
(43)		Edh at Noon	41	39	36	33	29	27	29	38	51	47	43	42	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2124	1922	1539	1211	976	885	1067	1378	1665	1858	2092	2108	(44)	
(45)		RadStd	159	136	154	138	124	111	87	100	154	182	188	130	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	+0.90	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	+1.48	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page